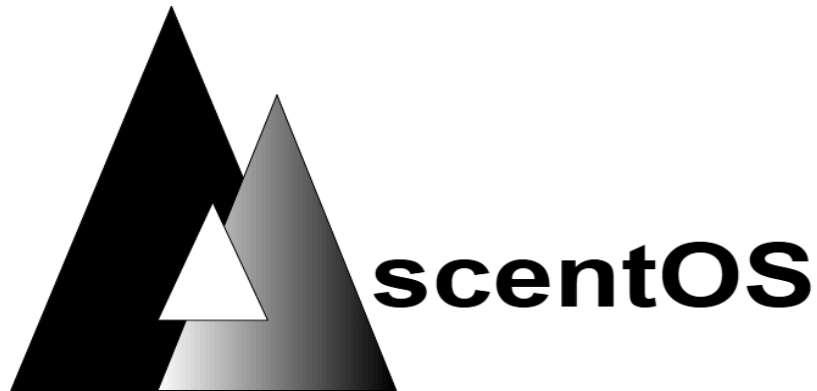




User Manual

Introducing AscentOS, our command line interface operating system project. Inside you shall find all the available commands, developed to help you interact with our operating system.

To use a command line interface, write any of the commands down below into the terminal, accompanied by any arguments the command requires, if any. All arguments are separated by spaces. If formatted correctly, the command will run and the consequences will be displayed.

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Commands

help

Arguments: [Name of Command] (*optional*)

Description:

Type in help to display every command available to the user.
Type a command's name after help to get more information on that command. Arguments surrounded by square brackets "[arg]" indicate an optional argument, and angled brackets "<arg>" indicate a required argument.

```
AscentOS >help
Command Information [Arguments are written in brackets, do not include brackets]:
----- System Utilites -----
help [cmd] - Shows available commands, if additional command is provided, shows a detailed description
version - Shows version and time of compilation
shutdown - Shuts down OS after confirmation
clear - Clears the terminal window of text
get_date - Shows OS date
set_date <date> - Sets the OS date to the date provided using mm/dd/yyyy format
get_time - Shows OS time
set_time <time> - Sets the OS time to the time provided using hh:mm:ss format
----- PCB Management -----
For PCB commands, you may omit '_pcb'
create_pcb <name> <class> <priority> - Creates a PCB with the given properties
delete_pcb <name> - Deletes the PCB with the given name
block_pcb <name> - Blocks the PCB with the given name
unblock_pcb <name> - Unblocks the PCB with the given name
suspend_pcb <name> - Suspends the PCB with the given name
resume_pcb <name> - Unsuspends the PCB with the given name
set_priority <name> <priority> - Modifies the priority of a PCB and moves it to the appropriate location
show_pcb <name> - Displays the pcb's name, priority, class, execution status,suspend status
show_ready - Displays the information of every pcb that is in the ready state
show_blocked - Displays the information of every pcb that is in the blocked state
show_all - Displays the information of every pcb
```

version/v

Arguments: None

Description:

Displays the version number R# alongside the compilation date.
Can alternatively be called by typing "v".

```
AscentOS >version
Version R2, Compilation Date: Feb 18 2026
AscentOS >v
Version R2, Compilation Date: Feb 18 2026
```

shutdown

Arguments: None

Description:

Shuts down the operating system after confirmation by the user.

```
AscentOS >shutdown
Are you sure you want to shutdown? Type y to confirm:
y
Shutdown confirmed. Exiting AscentOS...
klogv: Starting system shutdown procedure...
klogv: Halting CPU...
mp4@MasonsDell:/mnt/c/Users/might/Projects/Group-7$ |
```

get_date

Arguments: None

Description:

Returns the date in MM/DD/YYYY format after user execution.

```
AscentOS >get_date
02/19/2026
```

set_date

Arguments: <date> (required): MM/DD/YYYY

Description:

Takes the date written in the above format, where M is the month, D is the day, and Y the year, and sets it as the terminal date.

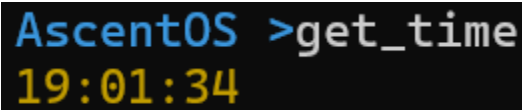
```
AscentOS >set_date 12/05/2022
Date set successfully!
AscentOS >get_date
12/05/2022
```

get_time

Arguments: None

Description:

Returns the current time set initially by the CPU in a 24 hour clock, military time, format.



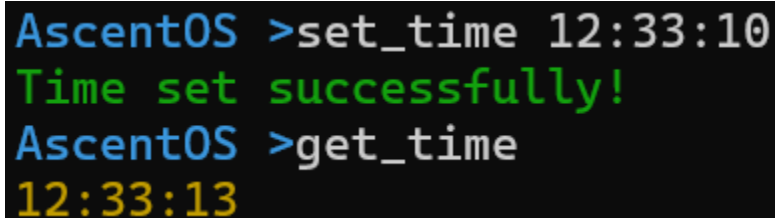
```
AscentOS >get_time
19:01:34
```

set_time

Arguments: <time>(*required*): HH:MM:SS

Description:

Takes the time written in the above format, where H is the hour, M is the minute, and S is the second, and sets it as the current terminal time.



```
AscentOS >set_time 12:33:10
Time set successfully!
AscentOS >get_time
12:33:13
```

clear

Arguments: None

Description:

Clears the terminal window of text by moving the terminal window such that the cursor is at the top left corner, and erasing all text from the cursor down.



create_pcb/create

This command is no longer supported.

delete_pcb/delete

Arguments: <name>(*required*): The name of an existing PCB

Description:

Deletes the PCB with the provided name, removing it from whichever queue it is located. Cannot delete a system pcb.

```
AscentOS >delete_pcb pcb1
PCB deleted successfully!
AscentOS >delete pcb2
PCB deleted successfully!
AscentOS >delete fake
Error: PCB with given name does not exist.
```

block_pcb/block

Arguments: <name>(*required*): The name of an existing PCB

Description:

Blocks the PCB with the provided name, and moves it to the appropriate queue. Cannot block a system pcb.

```
AscentOS >create pcb1 user 7
PCB created successfully!
AscentOS >block pcb1
PCB blocked successfully!
AscentOS >block pcb1
Error: PCB already blocked.
```

unblock_pcb/unblock

Arguments: <name>(*required*): The name of an existing blocked PCB

Description:

Unblocks the PCB with the provided name, and moves it to the appropriate queue. Cannot unblock a system pcb.

```
AscentOS >unblock pcb1
PCB unblocked successfully!
```

suspend_pcb/suspend

Arguments: <name>(*required*): The name of an existing non-system PCB

Description:

Suspends the PCB with the provided name, and moves it to the appropriate queue. Cannot suspend a system pcb.

```
AscentOS >suspend pcb1
PCB suspended successfully!
AscentOS >suspend pcb1
Error: PCB already suspended.
```

```
AscentOS >suspend pcb2
Error: A system process cannot be suspended.
```

resume_pcb/resume

Arguments: **<name>**(*required*): The name of an existing suspended non-system PCB

Description:

Resumes (un-suspends) the PCB with the provided name, and moves it to the appropriate queue. Cannot resume a system pcb.

```
AscentOS >create proc1 user 2
PCB created successfully!
AscentOS >suspend proc1
PCB suspended successfully!
AscentOS >resume proc1
PCB resumed successfully!
AscentOS >resume proc1
Error: PCB is not suspended.
```

set_priority

Arguments: **<name>**(*required*): The name of an existing PCB
<priority>(*required*): A number from 0 (highest priority) to 9 (lowest priority)

Description:

Sets the priority of the PCB with the provided name to the provided priority level. Cannot set the priority of a system pcb.

```
AscentOS >create pcb1 user 0
PCB created successfully!
AscentOS >set_priority pcb1 NAN
Error: Please enter an integer for priority.
AscentOS >set_priority pcb1 1
PCB priority set successfully!
```

show_pcb/show

Arguments: <name>(*required*): The name of an existing PCB

Description:

Shows information about the PCB with the provided name, such as its name, class, priority number, execution status, and suspend status.

```
AscentOS >create :) user 1
PCB created successfully!
AscentOS >show :)
PCB found:
  Name: :)
  Class: USER
  Priority: 1
  Execution Status: READY
  Suspend Status: NOT SUSPENDED
```

show_ready

Arguments: None

Description:

Shows the name, class, priority, execution status, and suspended status about each PCB in the Ready queue and Ready-Suspended queue.


```
AscentOS >show_ready
All Ready PCBs:
  Name: :)
  Class: USER
  Priority: 1
  Execution Status: READY
  Suspend Status: NOT SUSPENDED

All Ready-Suspended PCBs:
  Name: :P
  Class: USER
  Priority: 2
  Execution Status: READY
  Suspend Status: SUSPENDED
```

show_blocked

Arguments: None

Description:

Shows the name, class, priority, execution status, and suspend status about each PCB in the Blocked queue and Blocked Suspended queue.

```
AscentOS >show_blocked
All Blocked PCBs:
  Name: :D
  Class: USER
  Priority: 6
  Execution Status: BLOCKED
  Suspend Status: NOT SUSPENDED

All Blocked-Suspended PCBs:
  Name: :(
  Class: USER
  Priority: 9
  Execution Status: BLOCKED
  Suspend Status: SUSPENDED
```

show_all

Arguments: None

Description:

Shows the name, class, priority, execution status, and suspend status of all PCBs. PCBs are separated into ready, ready suspended, blocked, and blocked suspended queues.

```
AscentOS >show_all
All Ready PCBs:
  Name: :)
  Class: USER
  Priority: 1
  Execution Status: READY
  Suspend Status: NOT SUSPENDED

All Ready-Suspended PCBs:
  Name: :P
  Class: USER
  Priority: 2
  Execution Status: READY
  Suspend Status: SUSPENDED

All Blocked PCBs:
  Name: :D
  Class: USER
  Priority: 6
  Execution Status: BLOCKED
  Suspend Status: NOT SUSPENDED

All Blocked-Suspended PCBs:
  Name: :(
  Class: USER
  Priority: 9
  Execution Status: BLOCKED
  Suspend Status: SUSPENDED
```

load_r3

Arguments: <disp_state>(*required*): ready or suspended

Description:

Creates processes 1-5 and places them in either the ready queue or suspended queue based on the argument provided by the user.

```
AscentOS >load_r3 ready
5 out of 5 processes created sucessfully
```

```
AscentOS >load_r3 suspended
5 out of 5 processes created sucessfully
```

load_process

Arguments: <name>(*required*): must be process name process1-5

Description:

Creates processes 1-5 and places the created process in the ready queue.

```
AscentOS >load_process process1
Process successfully created!
```

resume_all

Arguments: None

Description:

Moves all processes from the ready suspended and blocked suspended queues to the ready and blocked queues

```
AscentOS >resume_all
Processes successfully resumed!
```

yield

This command is no longer supported.

alarm

Arguments: `<time>` (*required*): A time in HH:MM:SS format

`<message>` (*prompted*): A message to accompany the alarm

Description:

Creates an alarm process that will alert the user when the OS clock reaches the provided time. The user can also attach a message to the alarm. That will display when the alarm goes off.

```
AscentOS >get_time
IDLE PROCESS EXECUTING.
17:59:22
AscentOS >alarm 18:00:00
IDLE PROCESS EXECUTING.
Input a message to accompany alarm: Hello, World!
Alarm set successfully!
AscentOS >get_time
IDLE PROCESS EXECUTING.
17:59:59
AscentOS >
AscentOS >get_time
Alarm Is Ringing: Hello, World!
IDLE PROCESS EXECUTING.
18:00:06
AscentOS >
```

allocate_mem

This command is no longer supported.

Free_mem

This command is no longer supported.

show_allocated

Arguments: None

Description:

Displays all of the currently allocated memory blocks, showing both their starting address in hexadecimal and their size in decimal.

```
AscentOS >show_allocated
Allocated MCBs:

Start Address:0xD00088C
Size of Block: 6767

Start Address:0xD000450
Size of Block: 1064

Start Address:0xD000014
Size of Block: 1064
```

show_free

Arguments: None

Description:

Displays all of the currently free memory blocks, showing both their starting address in hexadecimal and their size in decimal.

```
AscentOS >show_free
Free MCBs:

Start Address:0xD00088C
Size of Block: 6767

Start Address:0x0
Size of Block: 41025
```